Agreement and adjudication procedure

Supports §5.5 (Who labelled) and Appendix E. The procedure for measuring agreement between the two coders and for settling disagreements, for all coding in Parts A–C. It was written before coding; its "Changes" table records the one change, made on 25 October 2026 before either coder had seen any disagreement. The second coder was the third author, so the fallback the procedure describes for the case that no second coder could be found (an AI judge panel) was not used.

"Coder 1" is the first author. "Pass 1" is the labelling of the live answers participants saw, "pass 2" the blind labelling of the offline comparison. "Pre-registration" means the study plan approved before the first session; it was not registered publicly (§5). Tracker identifiers, plan sections and file paths refer to the authors' project records.

Edits for anonymity: the authors' names became "the first author" (8 places) and "the third author" (2 places), and four pronouns referring to them became neutral.

Agreement and adjudication procedure (all coding in Parts A, B and C)

Who codes what (one second person, about 9 hours)

The first author is **coder 1 for everything**. A second coder is a different person, because agreement measured within the first author's own coding is intra-rater, and a reviewer reads it as a single-coder study. Recruiting anyone for the full ~25 hours looked unlikely, so the second person's share is cut to the material reviewers check hardest:

Material	Coder 1 (the first author)	Second coder (~9 h)	Blind to
Key check (12 study forms)	**done by the first author, by hand, against the live stores** — the keys were computed by the audit script, which reads the TTL files, so a hand check against the live graph and databases is independent of it	—	—
Part A validation (400 rows)	all	**seeded 100** (25 per stage; `draw_double_coding_sample.py --strata stage --fraction 0.25`)	automated labels; each other
Fresh vs corpus (≈90 + 90)	all	seeded 60	source; each other
Stage 5 Trust/Success (59 + 59)	all	— (coder 1 only; reported as single-coder)	old keyword labels
Pass 1: OntoSage answers participants saw	all (daily)	seeded **30 answers**, stratified by situation	the participant's decision
Pass 2: blind comparison (200)	all	seeded **100** (50 questions × 2 systems, both answers of a question together)	system identity; panel; each other

Material	Coder 1 (the first author)	Second coder (~9 h)	Blind to
Pass 2: A-minus answers (100, shuffled into the same blind set)	all	seeded **30**	system identity; panel; each other
Part C reasons	all	seeded 20%	—

Approximate time for the second person: Part A 100 rows ≈ 2.5 h; fresh-vs-corpus 60 ≈ 1 h; pass 1 30 answers ≈ 1 h; pass 2 100 answers ≈ 2.5 h; A-minus 30 answers ≈ 45 min; reasons ≈ 1 h; plus a 30-minute briefing on the codebooks — about 9 hours in all.

If no second person is confirmed before coding starts (the first author sets that date in their own plan; pre-declared fallback, registered in the pre-registration):

1. The second rater for Part A and pass 2 becomes an **AI judge panel**: three judges, each a model from a different family where available, never the systems' own `gpt-oss:20b`, majority label, with Fleiss' κ among the judges reported. The same codebook and rubric text go in the judge prompt, verbatim.
2. Report **human–panel agreement**, labelled as such, never as human inter-rater reliability.
3. The first author re-codes a seeded 10% of their own Part A and pass-2 labels **one week later**, blind to the first labels. Report this as intra-rater agreement.
4. State the limitation in the paper: the labels rest on one human coder.

Independence rules

1. Code in **separate copies** of the sheet; never open the other coder's file until both are locked (file hash recorded in `coding_log.csv`).
2. No discussion of specific rows before both sheets are locked. Questions about the codebook go into a written query log, answered for both coders at once.
3. The author's pass-1 labels on the answers sampled into pass 2 are compared with the author's own blind pass-2 labels. This checks drift in unblinded scoring.

Statistics

Field type	Statistic	Also report
Single-label (F4, F5, F6, W/WL/U/FD/OT)	Cohen's κ with 95% CI (bootstrap, 2,000 resamples)	raw agreement; confusion table
Multi-label (F1, F2, U sub-types, reason codes)	per-label κ; mean Jaccard overlap of label sets	raw agreement per label; prevalence
Ordinal (F7 certainty)	weighted κ (linear)	—

Interpretation is stated as numbers, not adjectives. If $\kappa < 0.60$ on a primary field, report it and run the headline analysis both on the adjudicated labels and on each coder's labels.

Adjudication

1. List disagreements (`adjudication_log_TEMPLATE.csv`), with `discuss`-flagged rows first.
2. Both coders meet; for each disagreement, record the agreed code and a one-line reason.
3. Both original rows are **kept**. The agreed row is added; nothing is overwritten.
4. If they still disagree, the supervisor decides, and that is recorded. *(Replaced 25 Oct 2026: see "Changes" below.)*
5. If a codebook rule is found ambiguous during adjudication, the clarification is written as a **note** in the codebook's change table. Labels already made are not re-coded unless both coders agree and every affected row is re-coded (and reported).

Files

- `sheets/*_TEMPLATE.csv`: copy to `<name>_<coder>.csv`, one per coder.
- `coding_log.csv`: date, coder, sheet, rows done, file SHA-256 at lock.
- Agreement scripts: `../F_Analysis/` (to be written against these sheet layouts; tested on synthetic labels first, tracker G4).

Changes

Date	Change	Why
25 Oct 2026	The second coder is the third author, the supervisor. Adjudication step 4 becomes: an item the two coders still disagree on after discussion keeps coder 1's label (the first author labelled every item); the paper reports how many items stayed disputed and gives the affected results on the second coder's labels as well.	The supervisor is now one of the two coders, so "the supervisor decides" would let one coder settle that coder's own disagreements. Decided by the first author and sent to the third author with the pack, before either coder had seen any disagreement.